Original Article

Cytogenetic and Genome Research

Cytogenet Genome Res
DOI: 10.1159/000444873

Accepted: January 14, 2016
by M. Schmid
Published online: May 13, 2016

Plantago lagopus B Chromosome Is Enriched in 5S rDNA-Derived Satellite DNA

Katrin Kumke[a] Jiří Macas[b] Jörg Fuchs[a] Lothar Altschmied[a] Jasmeet Kour[c] Manoj K. Dhar[c] Andreas Houben[a]

[a]Leibniz Institute of Plant Genetics and Crop Plant Research (IPK) Gatersleben, Stadt Seeland, Germany; [b]Biology Center CAS, Institute of Plant Molecular Biology, České Budějovice, Czech Republic; [c]School of Biotechnology, University of Jammu, Jammu, India

Key Words
Polymorphic A chromosome segment · Satellite repeat · Supernumerary chromosome · 5S rDNA

Abstract
B chromosomes are supernumerary dispensable parts of the karyotype which appear in some individuals of some populations in some species. Using advanced sequencing technology, we in silico characterized the high-copy DNA composition of *Plantago lagopus* with and without B chromosomes. The nuclear genome (2.46 pg/2C) was found to be relatively rich in repetitive sequences, with highly and moderately repeated elements making up 68% of the genome. Besides a centromere-specific marker, we identified a B-specific satellite and a repeat enriched in polymorphic A chromosome segments. The B-specific tandem repeat PLsatB originated from sequence amplification including 5S rDNA fragments.

© 2016 S. Karger AG, Basel

B chromosomes (Bs) are supernumerary dispensable parts of the genome. They occur in a wide range of taxa from fungi to plants and animals [reviewed in Jones and Rees, 1982]. Depending on the species, they may vary in size, structure and chromatin properties, but nevertheless they share certain characteristics which make them unique and distinguishable from other types of chromosome polymorphisms like for example aneuploidy. The question of the origin of B chromosomes arose with the first discovery of these supernumeraries and still remains unanswered for the majority of Bs. Generally, it is assumed that Bs are derived from standard chromosomes, either from the same (intraspecific origin) or from a related species (interspecific hybrid origin). The available experimental data support both scenarios, suggesting that their formation differs between different species and B chromosome types [reviewed in Houben et al., 2014].

Recent studies in rye [Martis et al., 2012] and in the fish species *Astatotilapia latifasciata* [Valente et al., 2014] using next-generation sequencing and sophisticated bioinformatics tools showed that in these species Bs had originated from multiple As, most likely by a pathway involving partial genome duplication, chromosome rearrangements, accumulation of organellar-derived DNA, and sequence amplification. Similarly, analysis of evolutionarily conserved A chromosome segments in wild canid species identified several regions of domestic dog A chromosome sequences that share sequence similarity with canid B chromosomes [Becker et al., 2011].

KARGER

E-Mail karger@karger.com
www.karger.com/cgr

© 2016 S. Karger AG, Basel
1424–8581/16/0000–0000$39.50/0

Andreas Houben
Leibniz Institute of Plant Genetics and Crop Plant Research (IPK) Gatersleben
Corrensstrasse 3
DE–06466 Stadt Seeland (Germany)
E-Mail houben@ipk-gatersleben.de

Downloaded by:
Zentralinstitut f. Pflanzengenetik
194.94.136.158 - 5/13/2016 8:32:10 AM

The Plantaginaceae *Plantago lagopus* is a weed of the Mediterranean region. A spontaneous trisomic plant (2n = 2x = 13, Triplo 4) of *P. lagopus* was recovered in 1982 in an experimental population [Sharma and Koul, 1984]. The trisomic plant was likely a product of a nondisjunction event, a feature observed in ~3% of the pollen mother cells of normal euploid plants. Crossing of these trisomic plants with euploids revealed a remarkable plasticity of the genome, and plants with 13 to 28 chromosomes were recovered. In one of the aneuploids, initially a primary trisomic for chromosome 2, the extra chromosome underwent rapid changes resulting in the formation of ring chromosomes and chromosome fragments. Finally the extra chromosome got stabilized as the proto-supernumerary isochromosome. Based on fluorescence in situ hybridization (FISH) and other molecular techniques, Dhar et al. [2002] proposed that the B chromosome arose as a result of massive amplification of sequences with similarities to 5S and 45S rDNA as well as to other sequences. To characterize the high-copy DNA fraction of this species, next-generation sequencing of plants without and with 1 B chromosome was carried out. In order to analyze the sequence composition of plants with and without Bs, we in silico compared abundant repeats based on next-generation sequence reads. Besides a centromere-specific marker, we identified a B-specific satellite and a repeat enriched in polymorphic A chromosome segments.

Materials and Methods

Plant Material

Plants of *P. lagopus* were raised in the green house, and the plants bearing B chromosomes were identified using Feulgen staining.

Flow Cytometric Genome Size Estimation

The genome size of *P. lagopus* was determined using a FACSAria Flow Sorter (BD Biosciences) following the protocol of Dolezel et al. [2007] using the nuclei isolation buffer according to Galbraith et al. [1983] supplemented with 50 µg/ml propidium iodide and 50 µg/ml DNase-free RNase. Either *Raphanus sativus* 'Voran' (IPK genebank accession number RA 34; 2C = 1.11 pg [Schmidt-Lebuhn et al., 2010]) or *Pisum sativum* 'Viktoria, Kifejtö Borsó' (IPK genebank accession number PIS 630; 2C = 9.09 pg [Dolezel et al., 2007]) were used as internal reference standard.

Illumina Sequencing of Genomic DNA

Genomic DNA of plants with 1 B (1B) and without B (0B) chromosome was used for Illumina sequencing as recommended in the TruSeqDNA v2 kit (Illumina). In brief, genomic DNA was fragmented by ultrasound (Covaris), blunted, adenylated, ligated with Illumina Y-shaped adapters, and amplified by PCR. The part of the genomic library with insert sizes between 300 and 400 bp was excised from a preparative agarose gel, the fragment length distribution verified on a Bioanalyzer chip (Agilent), the library quantified via qPCR and sequenced on one lane of an Illumina HiSeq2000 flowcell to produce 101-nt paired-end reads. The sequence reads are stored under PRJEB11733 (ENA/EBI).

Repeat Identification Using Similarity-Based Read Clustering

Repeat identification by similarity-based clustering of Illumina reads was performed using the RepeatExplorer pipeline [Novak et al., 2013] which was run on a Debian Linux server with 32 CPU cores and 64 GB RAM. The pipeline employs graph representation of read similarities to identify clusters of frequently overlapping reads representing various repetitive elements or their parts [Novak et al., 2010]. In addition, it provides information about repeat quantities (estimated from numbers of reads in the clusters), information about cluster connections via paired-end reads used to identify repeats split between multiple clusters, and outputs from BLASTn and BLASTx [Altschul et al., 1990] similarity searches to our custom databases of repetitive elements and repeat-encoded conserved protein domains that aid in repeat annotation. In order to reveal repeats accumulated on B chromosomes, the analysis was run in a comparative mode, performing simultaneous clustering of reads from 1B and 0B samples [Martis et al., 2012]. The structure of the cluster graphs was investigated using SeqGrapheR program [Novak et al., 2010].

Fluorescence in situ Hybridization

PLsatB and Pl-Cl1 probes were ordered as 5′-Alexa 488 or 5′-Cy3-labeled oligonucleotides (Eurofins MWG Operon; http://www.eurofinsdna.com) with the following sequences: PLsatB probe-A: 5′-TCA TCA TCA CAT TAT TCT CGG AGG ACC GCA AAT GTT TTT GGT AAA ATA CA-3′, probe-B: 5′-TCA TCG CCA CAG TAT TCG TGG TGG ATA ACA AAA ATT GTT ATA AAA TAC AT-3′, probe-C: 5′-TCA TTG CCA CTA TAT TCA TGG AGG ATA ACA AAA AGT GTT ACA AAA GAC AA-3′ (see fig. 2 for the probe positions and similarities) and Pl-Cl1 5′-CCT TAC TAC AAG TAG AAG TGC ATT CAC ACT CTA ATA ACT AGC TCT CTA AC-3′. To generate a CL94c29-specific probe, amplicons were labeled after PCR using the primers CL94c29_F1: 5′-GCT ACT GCC GGT CTT CTA CG-3′ and CL94c29_R1: 5′-AAT TGT CTC AGG GCC AGA TG-3′. The 5S rDNA probe was PCR generated from genomic DNA using the primers 5S rDNA-F: 5′-GGA TGC GAT CAT ACC AGC AC-3′ and 5S rDNA-R: 5′-GAC ATG CAA CTA TCT ATT TGT-3′. PCR amplicons were labeled by nick translation with Texas Red-, Cy5- or Alexa 488-dUTP. Chromosome preparation and FISH was done according to Ma et al. [2010]. Imaging was performed by using an Olympus BX61 microscope and an ORCAER CCD camera (Hamamatsu). All images were collected in gray scale and pseudocolored with Adobe Photoshop (Adobe).

Results

To investigate the repeat composition of *P. lagopus* B chromosomes, genomic shotgun sequencing of 2 genotypes differing in the presence of B chromosomes (0B and 1B) was performed using an Illumina platform. A ran-

Downloaded by:
Zentralinstitut f. Pflanzengenetik
194.94.136.158 - 5/13/2016 8:32:10 AM

Table 1. Composition of highly and moderately repeated sequences in *P. lagopus* plants lacking (0B) and carrying a B chromosome (1B)

Repeat	Genome proportion, %	
	0B	1B
LTR/*copia*		
Maximus/SIRE	26.70	25.34
TAR	0.99	0.95
Angela	0.53	0.52
Tork	0.26	0.29
AleII	0.16	0.14
AleI/Retrofit	0.04	0.04
LTR/gypsy		
Athila	11.14	10.54
Tat	4.47	4.25
Chromovirus	2.62	2.39
LTR/unclass.	0.80	0.82
LINE	0.20	0.18
Helitrons	1.00	0.97
DNA		
CACTA	2.71	2.58
Mutator	0.95	0.93
Satellites	3.55	6.70
rDNA	1.00	0.88
Unclassified	10.42	10.30
Total of genome	67.52	67.82

Repeat proportions were calculated by annotating clusters representing at least 0.01% of the genome.

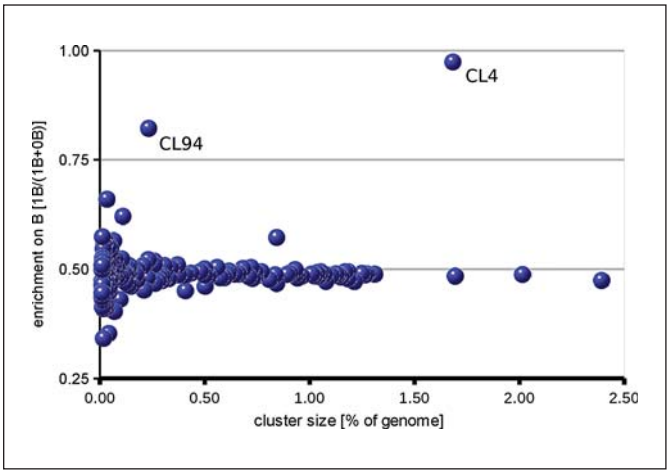


Fig. 1. Identification of repeat clusters enriched in *P. lagopus* genotype containing B chromosomes. The enrichment is revealed by calculating the ratio of reads derived from 1B genotype to the total number of reads (1B + 0B) in the cluster. The clusters CL4 (PLsatB) and CL94 with the highest enrichment are indicated.

domly sampled fraction of 1.2 million reads (representing 0.1× coverage of the genome assuming an estimated genome size of 2.46 pg/2C for the 0B genotype) were used for comparative clustering analysis, resulting in the identification of groups of frequently overlapping reads (clusters), each representing a single repeat family or its part. Since the number of reads in clusters is proportional to the genomic abundance of the corresponding repeats, this information was used to estimate genome proportions of all highly and moderately repeated sequences which were represented by 206 clusters with abundance exceeding 0.01% of the genome (table 1). The *P. lagopus* genome was found to be relatively rich in repetitive sequences, with highly and moderately repeated elements making up 68% of the genome. The majority of these sequences were classified into established groups of repetitive elements, revealing the Maximus/SIRE lineage of Ty1/*copia* LTR-retrotransposons to be the dominant repeat representing more than 25% of the genome. Except for the LTR/gypsy *Athila* lineage (11%), proportions of all other repeat types including other groups of LTR retrotransposons and DNA transposons did not exceed a few percent of the genome (table 1). In the B chromosome-containing genotype, an accumulation of satellite sequences (6.7%) compared to 0B plants (3.5%) was observed. The relative enrichment of the 2 repeat clusters, CL4 and CL94, identified as enriched in the plant possessing 1 B chromosome is shown in figure 1.

The cluster CL4 represented a satellite repeat designated PLsatB that makes up 3.3% of the 1B genotype but only 0.09% of the 0B genotype. Further investigation of read similarities employing cluster graph representation and sequence variants identified in assembled contigs revealed that there are at least 3 subfamilies of repeats differing in their sequences and monomer lengths (459–505 bp) (fig. 2a). It was also found that the satellite contains 2 regions (54 bp and 28 bp long) with sequence similarity to a genic and an intergenic spacer fragment of *P. lagopus* 5S rDNA (fig. 2b). Therefore, it is likely that the PLsatB repeat evolved from 5S rDNA sequences.

The repeat represented by the cluster CL94 was much less abundant compared to PLsatB and showed also lower enrichment in the analyzed 1B genotype. Cluster graph representation revealed a complex structure of this repeat, consisting of parts that were enriched in 1B reads while the rest of the graph contained reads from both

Downloaded by:
Zentralinstitut f. Pflanzengenetik
194.94.136.158 - 5/13/2016 8:32:10 AM

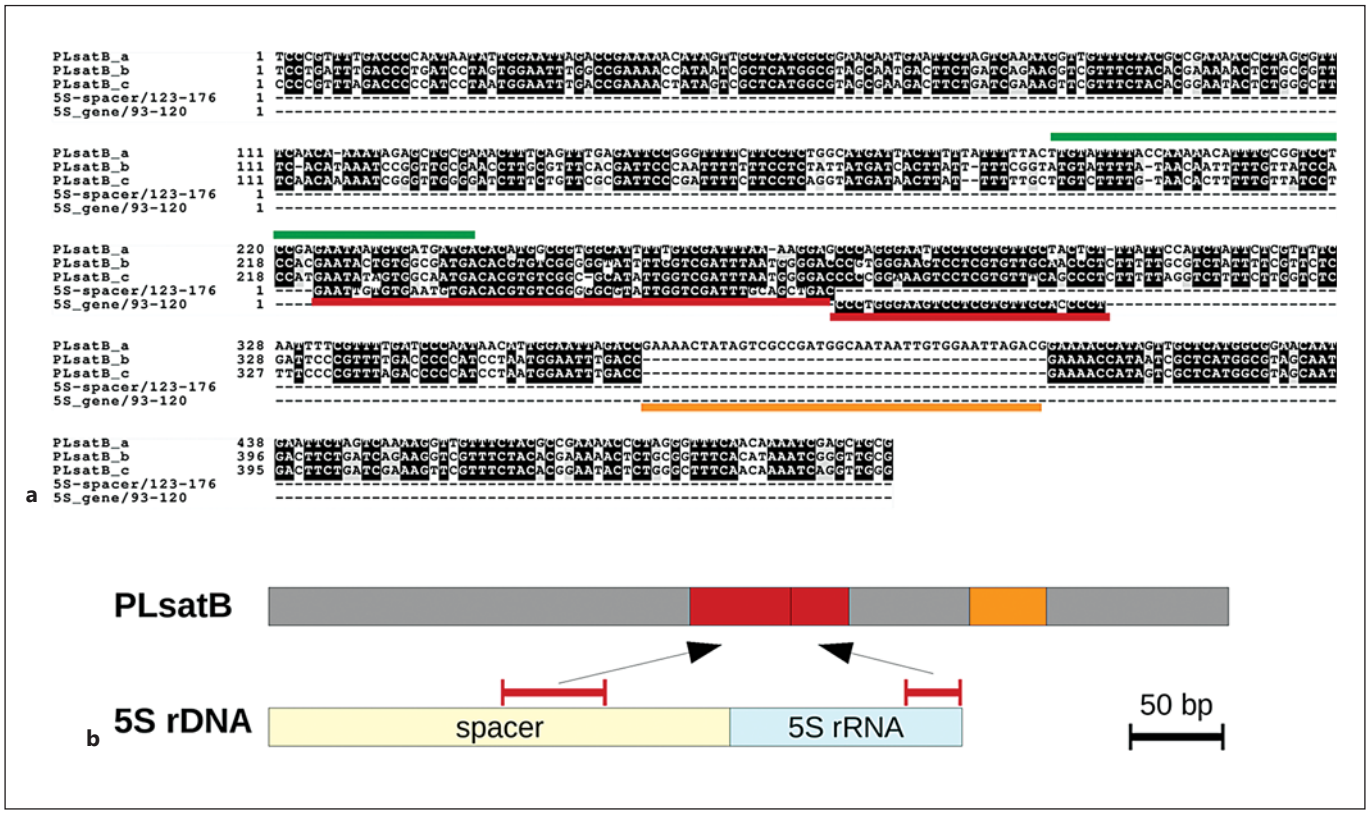


Fig. 2. Sequence composition of the PLsatB satellite and its similarity to 5S rRNA genes. **a** Multiple sequence alignment of 3 major variants of the PLsatB monomer and parts of the 5S rRNA gene and spacer sequences with similarities to PLsatB. Sequence regions with similarity to 5S rRNA genes are marked by red and position of hybridization probes by green lines. The orange line marks an indel in PLsatB monomers. **b** Schematic representation of the PLsatB monomer structure and its relation to the 5S rDNA cluster. Red and orange regions correspond to those marked on panel **a**.

samples. The B-enriched part of the graph had a circular structure, suggesting a tandem repeat. In addition to those B genotype-enriched repeats, we identified several other satellite repeats that likely locate on A chromosomes as potential cytogenetic markers. Among those we identified PLsat1, a tandem repeat with a monomer of 168 bp which makes up 2.3–2.5% of the genome.

In situ hybridization with the 1B genotype-enriched satellite PLsatB using probe A revealed an almost uniform labeling of the B chromosome at meiotic metaphase I, while extended mitotic prometaphase chromosomes showed a more clustered distribution of the hybridization signals (fig. 3a, b). In any case, no signals were detectable on the A chromosomes. No obvious differences regarding the signal distribution of the other 2 PLsatB subfamilies represented by probes B and C were found (data not shown). In order to test whether the coding region of the 5S rDNA of *P. lagopus* cross-reacts with the B chromosome too, a PCR-generated and sequence-confirmed 5S rDNA probe was used in parallel. Only one of the A chromosome pairs showed 5S rDNA signals, and no signal was detected on the Bs (fig. 3). Hence, the in total 82-bp-long region of the 5S rDNA probe with similarity to PLsatB was not sufficient to label Bs by FISH.

PLsat1 revealed (peri-)centromere-associated signals for 5 of the 6 A chromosome pairs (fig. 3a–c). The B chromosomes displayed only very weak centromeric signals, which were only detectable after an extended exposure time. After hybridization with the second 1B genotype-enriched repeat Pl-Cl94, only 1 A chromosome displayed a specific signal near the centromere (fig. 3c). The corresponding homolog, other A chromosomes as well as the B chromosomes did not reveal any detectable signals (fig. 3d). Additionally, also plants without Pl-Cl94 signals were found indicating that Pl-Cl94 identifies a polymorphic A chromosome segment.

Downloaded by:
Zentralinstitut f. Pflanzengenetik
194.94.136.158 - 5/13/2016 8:32:10 AM

Discussion

What mechanism could account for the accumulation of a satellite with partial similarity to the 5S rDNA in the B chromosome of *P. lagopus*? The involvement of rDNA-derived repeats in the evolution of B chromosomes does not appear to be accidental, because rDNA loci have been detected on Bs of many species of both plants (e.g. *Crepis capillaris* [Maluszynska and Schweizer, 1989], *Brachycome dichromosomatica* [Donald et al., 1997]) and animals (e.g. *Rattus rattus* [Stitou et al., 2000], *Trichogramma kaykai* [van Vugt et al., 2005], for additional examples see the review of Green [1990]). Except for the 45S rRNA genes on the B chromosome of *C. capillaris*, there has been no direct molecular evidence demonstrating transcription of B chromosome-located rDNA in any plant species [Leach et al., 2005]. In certain cases, noncoding tandem repeats evolved from rDNA sequences. This phenomenon has been described for some plant species like *Vicia sativa* [Macas et al., 2003] and *Solanum lycopersicum* [Jo et al., 2009] that exhibit sequences of satellite DNA similar to the intergenic spacer of the 45S rDNA. Also in animals some cases have been reported, such as the fish *Hoplias malabaricus*, in which a satellite DNA with high similarity to 5S rDNA was found in the centromeric regions of various chromosomes, suggesting that this satellite DNA originates from the 5S rDNA [Martins et al., 2006]. Similarly, centromeric satellite DNA derived from the 5S rDNA, probably by gene duplication and nucleotide divergence, was reported for the amphibian *Physalaemus cuvieri* [Vittorazzi et al., 2011].

In addition to a B chromosome-specific satellite sequence, we identified a satellite (Pl-CI94) specific for a polymorphic A chromosome segment in *P. lagopus*. Intraspecific variations in size and number of chromosomal regions between homologous chromosomes have been observed in various animal and plant species [reviewed in Houben et al., 2000]. In some species such regions are also called 'supernumerary segments', e.g. *Rumex* [Wilby and Parker, 1988] and grasshopper [Hewitt and John, 1968], by analogy with supernumerary (B) chromosomes, with which they share several characteristics, i.e. they are often heterochromatic and dispensable.

The DNA composition of the polymorphic segment of *P. lagopus,* which is composed of a high-copy tandem repeat, is similar to the DNA composition of the polymorphic heterochromatic segments of maize [Peacock et al., 1981; Ananiev et al., 1998] and the Australian daisy *B. dichromosomatica* [Houben et al., 2000]. In both species, repeating units arranged in tandem arrays are mainly involved in the formation of knob heterochromatin. An evolutionary link between Bs and polymorphic A segments has been proposed for *B. dichromosomatica* as both share similar sequences [Houben et al., 2001]. How-

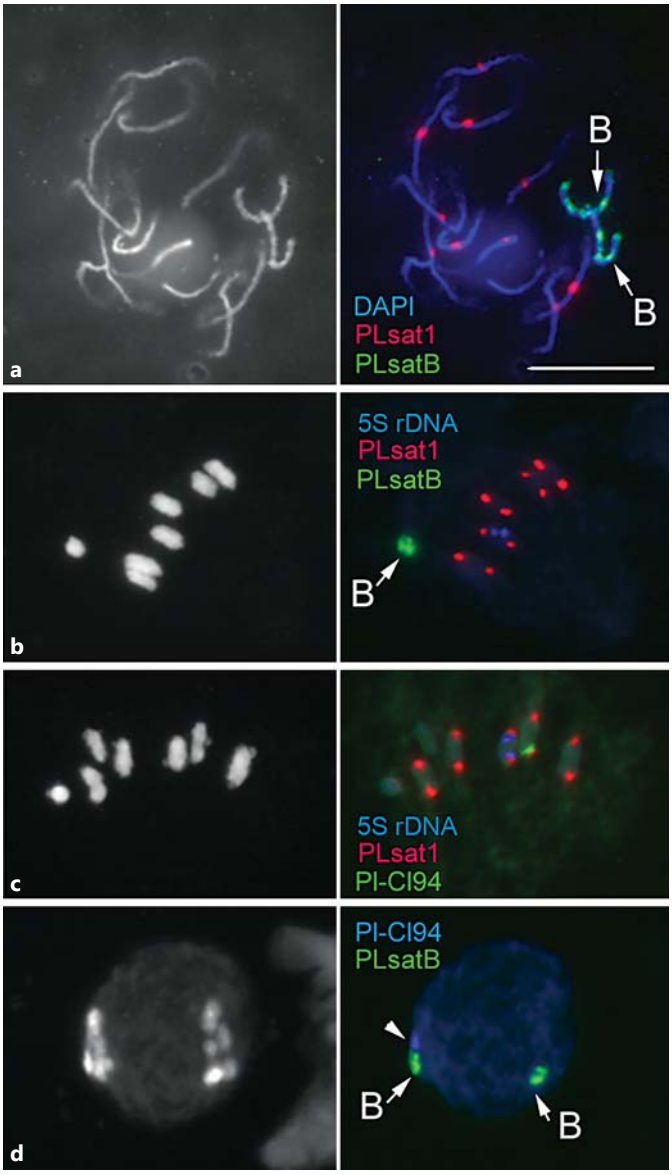


Fig. 3. FISH analysis of *P. lagopus* chromosomes (2n = 12 A chromosomes + Bs) of mitotic (**a**) and meiotic cells (**b–d**). The B-specific repeat PLsatB revealed either clustered (**a**) or an almost uniform (**d**, **c**) labeling exclusively of the B chromosomes. **a–c** The centromeres of 5 of the 6 A chromosome pairs are enriched in PLsat1 repeats. **b**, **c** Metaphase I; neither the B chromosome nor the Pl-Cl94-positive A chromosome shows 5S rDNA-specific signals. **d** Anaphase I; the Pl-Cl94-positive polymorphic A segment is indicated by an arrowhead. The Bs are indicated with arrows in **a**, **b** and **d**. Bar = 5 μm.

Downloaded by:
Zentralinstitut f. Pflanzengenetik
194.94.136.158 - 5/13/2016 8:32:10 AM

ever, no similarity between both dynamic components of the genome was found in *P. lagopus*. The question concerning the nature of the mechanisms resulting in the maintenance of the B chromosome and the A chromosome segment polymorphism in a natural population remains to be answered.

In summary, comparative in silico analysis of next-generation genomic DNA reads of genotypes with and without B chromosomes resulted in efficient identification of a B-specific repeat in *P. lagopus*. A major component of the B represents a tandem repeat originated by sequence amplification including 5S rDNA fragments.

Acknowledgements

We would like to thank D. Stengel (IPK, BIT) for submitting the sequence data. The work was supported by a Deutsche Forschungsgemeinschaft grant (DFG HO 1779/18-1) to A.H., the Czech Science Foundation grant (GBP501/12/G090) to J.M. and DST-DAAD to M.K.D. and A.H., DST and DBT (BIF), Government of India to M.K.D. and CSIR, New Delhi to J.K.

Disclosure Statement

The authors have no conflicts of interest to declare.

References

Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ: Basic local alignment search tool. J Mol Biol 215:403–410 (1990).

Ananiev EV, Phillips RL, Rines HW: Complex structure of knob DNA on maize chromosome 9. Retrotransposon invasion into heterochromatin. Genetics 149:2025–2037 (1998).

Becker SE, Thomas R, Trifonov VA, Wayne RK, Graphodatsky AS, Breen M: Anchoring the dog to its relatives reveals new evolutionary breakpoints across 11 species of the Canidae and provides new clues for the role of B chromosomes. Chromosome Res 19:685–708 (2011).

Dhar MK, Friebe B, Koul AK, Gill BS: Origin of an apparent B chromosome by mutation, chromosome fragmentation and specific DNA sequence amplification. Chromosoma 111:332–340 (2002).

Dolezel J, Greilhuber J, Suda J: Estimation of nuclear DNA content in plants using flow cytometry. Nat Protoc 2:2233–2244 (2007).

Donald TM, Houben A, Leach CR, Timmis JN: Ribosomal RNA genes specific to the B chromosomes in *Brachycome dichromosomatica* are not transcribed in leaf tissue. Genome 40:674–681 (1997).

Galbraith DW, Harkins KR, Maddox JM, Ayres NM, Sharma DP, Firoozabady E: Rapid flow cytometric analysis of the cell cycle in intact plant tissues. Science 220:1049–1051 (1983).

Green DM: Muller's ratchet and the evolution of supernumerary chromosomes. Genome 33:818–824 (1990).

Hewitt GM, John B: Parallel polymorphism for supernumerary segments in *Chorthippus parallelus* (Zetterstedt). I. British populations. Chromosoma 25:319–342 (1968).

Houben A, Wanner G, Hanson L, Verlin D, Leach CR, Timmis JN: Cloning and characterisation of polymorphic heterochromatic segments of *Brachycome dichromosomatica*. Chromosoma 109:206–213 (2000).

Houben A, Verlin D, Leach CR, Timmis JN: The genomic complexity of micro B chromosomes of *Brachycome dichromosomatica*. Chromosoma 110:451–459 (2001).

Houben A, Banaei-Moghaddam AM, Klemme S, Timmis JN: Evolution and biology of supernumerary B chromosomes. Cell Mol Life Sci 71:467–478 (2014).

Jo SH, Koo DH, Kim JF, Hur CG, Lee S, et al: Evolution of ribosomal DNA-derived satellite repeat in tomato genome. BMC Plant Biol 9:42 (2009).

Jones RN, Rees H: B Chromosomes, ed 1 (Academic Press, London 1982).

Leach CR, Houben A, Field B, Pistrick K, Demidov D, Timmis JN: Molecular evidence for transcription of genes on a B chromosome in *Crepis capillaris*. Genetics 171:269–278 (2005).

Ma L, Vu GTH, Schubert V, Watanabe K, Stein N, et al: Synteny between *Brachypodium distachyon* and *Hordeum vulgare* as revealed by FISH. Chromosome Res 18:841–850 (2010).

Macas J, Navrátilová A, Mészáros T: Sequence subfamilies of satellite repeats related to rDNA intergenic spacer are differentially amplified on *Vicia sativa* chromosomes. Chromosoma 112:152–158 (2003).

Maluszynska J, Schweizer D: Ribosomal RNA genes in B chromosomes of *Crepis capillaris* detected by non-radioactive in situ hybridization. Heredity 62:59–65 (1989).

Martins C, Ferreira IA, Oliveira C, Foresti F, Galetti PM Jr: A tandemly repetitive centromeric DNA sequence of the fish *Hoplias malabaricus* (Characiformes: Erythrinidae) is derived from 5S rDNA. Genetica 127:133–141 (2006).

Martis MM, Klemme S, Banaei-Moghaddam AM, Blattner FR, Macas J, et al: Selfish supernumerary chromosome reveals its origin as a mosaic of host genome and organellar sequences. Proc Natl Acad Sci USA 109:13343–13346 (2012).

Novak P, Neumann P, Macas J: Graph-based clustering and characterization of repetitive sequences in next-generation sequencing data. BMC Bioinformatics 11:378 (2010).

Novak P, Neumann P, Pech J, Steinhaisl J, Macas J: RepeatExplorer: a Galaxy-based web server for genome-wide characterization of eukaryotic repetitive elements from next-generation sequence reads. Bioinformatics 29:792–793 (2013).

Peacock WJ, Dennis ES, Rhoades MM, Pryor AJ: Highly repeated DNA sequence limited to knob heterochromatin in maize. Proc Natl Acad Sci USA 78:4490–4494 (1981).

Schmidt-Lebuhn AN, Fuchs J, Hertel D, Hirsch H, Toivonen J, Kessler M: An Andean radiation: polyploidy in the tree genus *Polylepis* (Rosaceae, Sanguisorbeae). Plant Biol (Stuttg) 12:917–926 (2010).

Sharma PK, Koul AK: Genetic diversity among Plantagos. III. Primary trisomy in *Plantago lagopus* L. Genetica 64:135–138 (1984).

Stitou S, Diaz de La Guardia R, Jiménez R, Burgos M: Inactive ribosomal cistrons are spread throughout the B chromosomes of *Rattus rattus* (Rodentia, Muridae). Implications for their origin and evolution. Chromosome Res 8:305–311 (2000).

Valente GT, Conte MA, Fantinatti BE, Cabral-de-Mello DC, Carvalho RF, et al: Origin and evolution of B chromosomes in the cichlid fish *Astatotilapia latifasciata* based on integrated genomic analyses. Mol Biol Evol 31:2061–2072 (2014).

van Vugt J, de Nooijer S, Stouthamer R, de Jong H: NOR activity and repeat sequences of the paternal sex ratio chromosome of the parasitoid wasp *Trichogramma kaykai*. Chromosoma 114:410–419 (2005).

Vittorazzi SE, Lourenco LB, Del-Grande ML, Recco-Pimentel SM: Satellite DNA derived from 5S rDNA in *Physalaemus cuvieri* (Anura, Leiuperidae). Cytogenet Genome Res 134:101–107 (2011).

Wilby AS, Parker JS: The supernumerary segment systems of *Rumex acetosa*. Heredity 60:109–117 (1988).

Downloaded by:
Zentralinstitut f. Pflanzengenetik
194.94.136.158 - 5/13/2016 8:32:10 AM